CHM_159
Worksheet_Chapter_2_Answers

nature of light, photoelectric effect, photon energy, electronic transitions in the H-atom, de Broglie wavelength, quantum numbers and atomic orbitals

Key equations

$c = \lambda\nu$ $E = h\nu$ $h\nu = W + KE$ $h\nu = h\nu_{th} + KE$ $KE = \frac{1}{2}mv^2$ $h = 6.63 \times 10^{-34}$ J

$$E_n = -2.18 \times 10^{-18}\ \text{J}\left(\frac{1}{n^2}\right) \qquad \Delta E = h\nu = -2.18 \times 10^{-18}\ \text{J}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{\lambda} = \frac{-2.18 \times 10^{-18}\ \text{J}}{hc}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) \qquad \lambda = \frac{h}{mu} \qquad c = 3.00 \times 10^8\ \text{m/s}$$

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} \qquad \Delta x \cdot m\Delta u \geq \frac{h}{4\pi} \qquad 1\ \text{J} = \text{kg. m}^2/\text{s}^2$$

wavelength ,frequency and energy

1. a. What is the frequency of light having a wavelength of 456 nm?
   b. What is the wavelength (in nm) of radiation having a frequency of 2.45×10^9 Hz? (This is the type of radiation used in microwave ovens.)
   c. Calculate the energy (in joules) of a single photon associated with a frequency of 7.5×10^{14} Hz

Ans : a. ν = = 6.58 × 10^{14} s^{-1} or 6.58 × 10^{14} Hz
b. λ = 1.22 x 10^8 nm
c. E = 4.97 x 10^{-19}J

2. Calculate the wavelenth , in meters, of microwave radiation that corresponds to an energy of 2.63 J/mol of photons.

Ans : 4.55 x 10^{-2} m

photoelectric effct

3. Light of wavelength 221.0 nm striking a sample of tungsten causes electrons to be ejected with a speed of 6.04×10^5 m/s. Calculate the work function(W) /binding energy of tungsten. ($E = W + \frac{1}{2}mv^2$, m= mass of electron = 9.10939×10^{-28} g, v = speed of electron).

Ans : 7.34 x 10^{-19}J

4. The work function of a given metal is 1.66x 10^{-17} J. This is the minimum energy that a photon must have in order to dislodge an electron from the metal. If the wavelength of the radiation incident on the metal is 8.8 x 10^{-9} m, what is the kinetic energy of the ejected electron.

Ans : 6.00 x 10^{-18}J

5. Calculate the wavelength(nm) of light that strikes a metal with a threshold frequency of 2.84 x $10^{14}s^{-1}$ and ejects an electron with a kinetic energy of 5.48 x 10^{-19}J.

Ans : 270. nm

Bohr Theory of Hydrogen atom

6. Use the Bohr equation to find the wavelength (in nm) of the photon emitted when an H atom undergoes a transition from n=5 to n=2.

Ans: 434 nm

7. How many lines would be in the emission spectrum of hydrogen if the hydrogen atom had only 10 energy levels?

Ans : 45

8. An electron in the hydrogen atom in an excited state with n = 7 drops to a lower energy level and emits a photon of wavelength 1006 nm. . What is the final energy level of the electron?

Ans : n=3

9. Consider the following energy levels of a hypothetical atom:

E_4: -1.0×10^{-19} J
E_3: -5.0×10^{-19} J
E_2: -1.0×10^{-18} J
E_1: -1.5×10^{-18} J

a. What is the wavelength of the photon needed for an electron to make a transition from E_1 to E_4?

b. What is the energy (in joules) a photon must have to excite an electron from E_2 to E_3?

c. When an electron drops from the E_3 level to the E_1 level, the atom is said to undergo emission. Calculate the wavelength of the photon emitted in this process.

Ans: a. $\lambda = 1.4 \times 10^2$ nm b. 5.0×10^{-19} J c. $\lambda = 2.0 \times 10^2$ nm

De Broglie and Heisenberg's Principle

10. Protons can be accelerated to speeds near that of light in particle accelerators. Estimate the wavelength (in nm) of such a proton moving at 2.90×10^{8} m/s (mass of a proton = 1.673×10^{-24} g).

Ans: 1.37×10^{-6} nm

11. a. The mass of a base-ball is 142g. Calculate the uncertainty in position if the velocity is 100.0 mi/h + 1.0 %. (1 mi = 1609 m). $\Delta x.m\Delta v \geq h/4\pi$.

b. The mass of an electron is 9.10939×10^{-31} kg. Calculate the uncertainty in position if the velocity is 100.0 mi/h $\pm$ 1.0 %. (1 mi = 1609 m)

Ans : a. $\Delta x = 8.30 \times 10^{-34}$ m
b. $\Delta x = 1.30 \times 10^{-4}$ m

quantum numbers

12. a .i What is the maximum number of electrons that could be designated by the quantum numbers n = 3, ℓ= 2?
ii. How many subshells are there in the shell designated by n = 4? Name them.
iii. What is the total number of orbitals in the shell designated by n = 5?
iv. What is the minimum value of the principal quantum number for an orbital in which m_ℓ = -3?

Ans : Answers: i) 10 electrons ii) 4s,4p,4d,4f iii) 25 orbitals iv) minimum of n = 4

b. Indicate whether the following sets of quantum numbers are permissible. If they are, state the orbital that they designate. If they are not, state which number(s) is/are not permissible.

i) n = 4, ℓ= 2, m_ℓ = 0, m_s = -½
ii) n = 3, ℓ= 3, m_ℓ = 2, m_s = -1
iii) n = 1, ℓ= 0, m_ℓ = 1, m_s = ½
iv) n = 6, ℓ= 0, m_ℓ = 0, m_s = -½
v) n = 2, ℓ= 1, m_ℓ = -2, m_s = -½

i) 4p ii) not permissible(l cannot be 3 if n=3, m_s cannot be -1) iii) not permissible(m_l cannot be 1 if l=0) iv) 6s
v) not permissible (m_l cannot be -2 if l=1)